

Div B: Experimental Design Report

Part 1 : Design and Construction of the Experiment

Time: 20 minutes

Competition: _____ Date: _____

School: _____ Team Number: B _____

Student: _____

Student: _____

Student: _____

FINAL SCORE

Team Kits: Each team may bring in:

- ☐ 1 linear measuring device
- ☐ 1 timing device
- ☐ 1 calculator of any kind
- ☐ 3 goggles

Tie Breaker Order

K. Analysis of Claim/Evidence/Reasoning
E. Procedure and Set-Up Diagrams
C. Variables
G. Data Table
H. Graph

Penalty Scores: Multiply to Total Score

Circle all that apply

<i>x .95</i>	Follow clean-up procedures before the end of the event
<i>x .95</i>	Use at least 2 of the provided materials
<i>x .25</i>	Collect data by experimenting on-site or falsified /made up data
<i>up to x .75</i>	Addressed the assigned question/topic area



Exploring the World of Science

**Blank sheet for event supervisor
to insert a prompt and any other
necessary background information.**

A Statement of Problem

B Hypothesis

C Variables

Independent Variable (IV)		
Define		Tools & Operationally defined
Level 1	Level 2	Level 3
Dependent Variable (DV)		
Define		Tools & Operationally defined
Control Variables (CV)		
Define		

D Materials

E Procedure and Set-up & Diagrams

F Qualitative Observations

Set-Up

Procedure

Results

Raw Table

Condensed Table

Div B Experimental Design Report

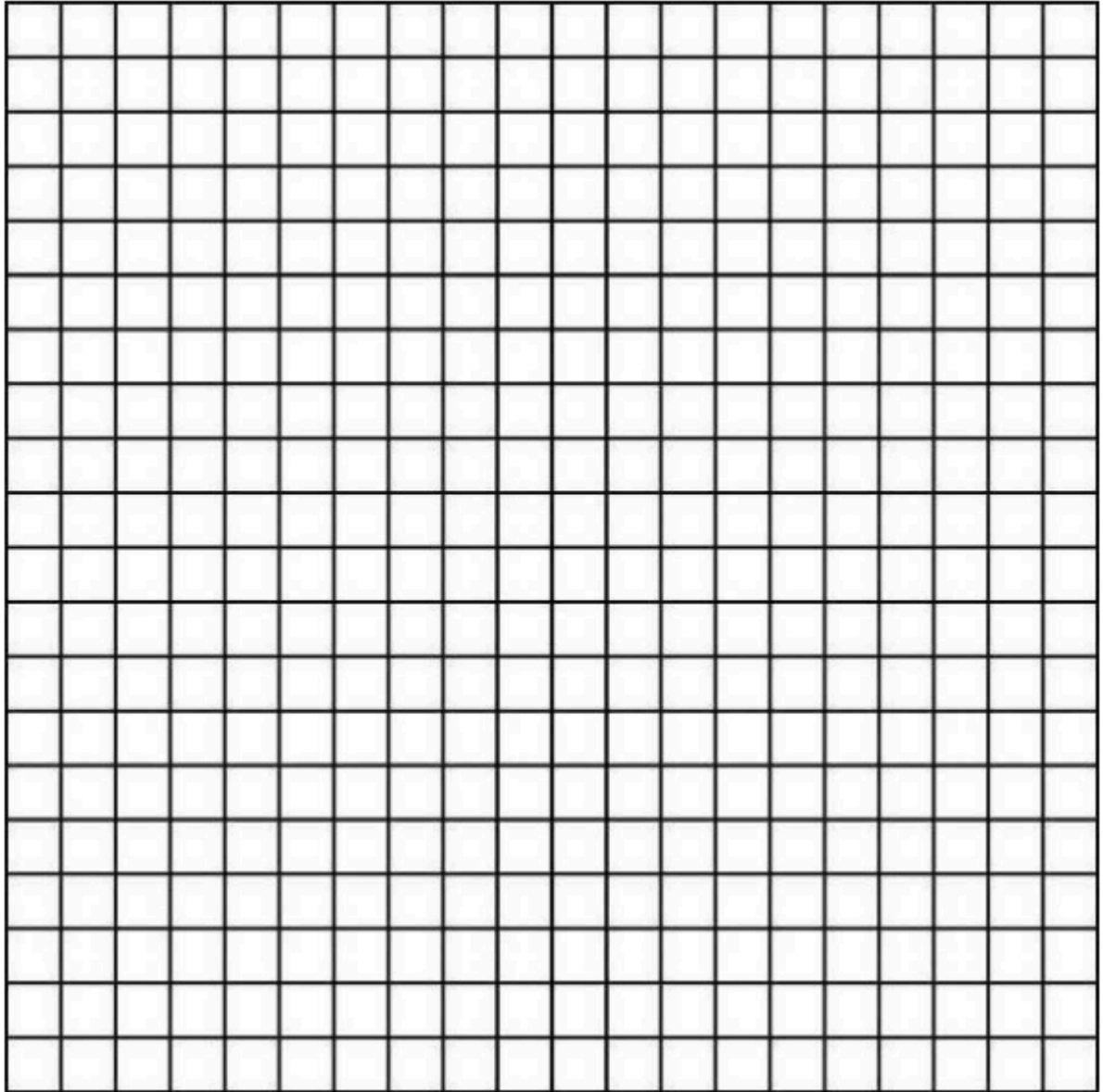
Part 2: Data, Analysis, and Conclusions

Time: 30 minutes

School: _____

Students may continue to experiment and work on Part 1 during this time.

H Graph



I Statistics

Central Tendency

Statistic #1

Statistic #2

Statistic #3

Variation

Statistic #1

Statistic #2

Statistic #3 (State/Nationals only)

Statistic #4 (State/Nationals only)

J Possible Experimental Errors

Specific Error Type <i>circle one</i>	Specific Error Identified	Specific Error in Results
Random Procedural Systematic		
Random Procedural Systematic		

K Analysis of Claim/Evidence/Reason

Data Trend

Claim

Evidence

Reasoning

Outliers

Claim

Evidence

Reasoning

K Analysis of Claim/Evidence/Reason - *continued*

Variation: State/Nationals only

Claim

Evidence

Reasoning

L Conclusion

Hypothesis Restated

Claim: Accept or Reject Hypothesis

Evidence

Reasoning

M Improvement, Applications & Recommendations for Further Use

Improvements for the Current Experiment

Practical Applications

Future Experiments
